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UGC & TREND ANALYZER

Sentiment, keywords and themes
pulled from Amazon product reviews.

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SATISFIED CUSTOMERS ARE NOT SILENT

Most systems that read customer feedback at scale attach the star rating to the whole review. This one scores every sentence on its own — and the gap between the two methods is where the useful information was hiding.

17.5%

OF SENTENCES INSIDE
5★ REVIEWS ARE COMPLAINTS

21.9%

OF ALL SENTENCES WERE
MISLABELLED BY RATING

88.2%

AUDITED THEME
ACCURACY

95.1%

AUDITED POLARITY
ACCURACY

What was built

A ten-stage pipeline over 4,173 Amazon Fine Food reviews — 9,954 sentences, 14 product categories, 2003–2012. Each sentence is embedded, scored for polarity by a review-tuned language model, and assigned to one or more of sixteen aspect themes by zero-shot entailment against written definitions. A review-level regression then ranks themes by their estimated drag on the star rating, and weighted log-odds identifies the vocabulary that separates praise from complaint in each category.

What it outputs

Per category, two decisions rather than a dashboard: **a ranked fix list** — ordered by how often a problem occurs there multiplied by what it is associated with costing, carrying 95% confidence intervals — and **the language to lead with**, drawn from the words customers themselves use when they are happy. Each category also gets a short written brief in which every single number is automatically checked against the underlying evidence before it is allowed to render.

The errors are not random. They concentrate exactly where a brand team would most want to look: in the complaints of customers who stayed.

THE CHEAPEST PROBLEMS TO FIX ARE INVISIBLE

A customer who gives five stars and mentions one broken thing is the most valuable signal a feedback system can receive. They have not churned. They are still talking. And the standard method throws that sentence away.

The mechanism of the error

When a review's rating is applied to its text, "the packaging arrived crushed" inside a four-star review is recorded as *positive evidence*. Every share-of-voice figure computed downstream inherits that mistake. Measured directly against sentence-level scoring, 21.9% of sentences carried the wrong label — and the misses are systematically concentrated in well-rated reviews, because that is where mixed feedback lives.

STAR RATING	SHARE OF ITS SENTENCES THAT ARE COMPLAINTS
5★	17.5% — invisible to a rating-based read
4★	26.5%
3★	55.4%
1★	79.2% — and 20.8% are <i>not</i> complaints

Volume is not importance

The second correction matters as much. Ranking themes by how often they are discussed sends teams after the loudest complaint rather than the costliest one. Here, themes are ranked by **expected drag** — how often a complaint occurs in that category, multiplied by the rating difference associated with it. Two themes were excluded from the ranking entirely because they restate the outcome rather than explain it.

One finding inverts the assumption completely: complaints about not being able to *find* a product travel with ratings 0.22★ **higher**. Scarcity complaints come from devotees. That is a distribution signal, not a defect.

ANY TEXT WITH A SCORE ATTACHED

Nothing about this method is specific to food, to Amazon, or to stars. It applies wherever an organisation collects free-text feedback alongside a numeric outcome — which is to say, almost everywhere. The rating-inheritance bug travels with the pattern.

SaaS support & CSAT

Ticket text inherits the CSAT score, so friction described inside "satisfied" tickets is never counted. Aspect extraction surfaces it; drag ranking orders it by association with the outcome you actually care about — renewal, not the survey.

App store reviews

A five-star review naming a missing feature is the highest-value product input a team can get, and it is exactly what a star-weighted dashboard discards. Aspect themes make "requested but absent" a trackable quantity.

Employee engagement

Free-text comments paired with an eNPS score, same inheritance problem. Because themes here are *aspects* rather than sentiments, "manager support" can be tracked as improving or worsening — which "negative sentiment is up 4%" never tells you.

Patient experience

Survey comments attached to an overall score. Complaints buried in high-scoring feedback — discharge instructions, waiting communication — are usually operational and cheap to fix, precisely because the patient was otherwise satisfied.

Financial services complaints

Regulated complaint narratives need triage, not sentiment. Aspect assignment with a published error rate and an explicit "unassigned" class is the difference between a model an auditor accepts and one they do not.

Hospitality & travel

The archetypal mixed review — "loved the hotel, the air conditioning was broken" — is a single sentence pair that rating-based reads collapse into one number. Multi-label assignment keeps both halves.

What a new domain actually needs

Free text, a numeric outcome, and enough volume to estimate effects — a few thousand responses is sufficient. The taxonomy is rewritten per domain and takes an afternoon; every stage after it is unchanged. The pipeline runs on a laptop GPU with no API costs.

WHAT IT GETS WRONG, STATED FIRST

A pipeline that assigns labels with a model owes its reader an error rate. This one publishes its own, including the two earlier versions that were not good enough.

ASSIGNMENT METHOD	THEME ACCURACY	WHY IT CHANGED
Clusters used directly as labels	56.2%	Clustering must place every sentence somewhere, so chatter lands inside a theme
Zero-shot entailment	82.4%	Compares each sentence against an explicit definition; can assign none
— pooling rule removed	88.2%	An over-aggressive rule had merged a valid theme into "Other"

Two methods were tried and rejected, and both are documented rather than quietly dropped. **HDBSCAN** — the conventional modern choice — returned two clusters and left 90% of sentences as noise; keeping it would have meant discarding nine tenths of the evidence to preserve a fashionable technique. **Feeding full theme definitions** to the entailment model left 62.5% of the corpus unassigned and erased "Price & value" from a corpus of food reviews; replacing them with short label phrases fixed it with no change of model or data.

Claims are checked before they are shown

Every number in every written brief is parsed and matched against the evidence behind it — 103 claims, none unmatched. It has caught real errors. It also has a blind spot worth naming: it verifies that a number exists, not that a *comparison* built on it is true. Two false superlatives passed it and were caught only by hand.

What this cannot tell you

Nothing causal. Theme polarity is extracted from the same text that produced the rating, so the association is mechanical by construction. These rankings are a triage order, not a treatment effect. **Nothing about individual products** — 3,438 products across 4,173 reviews is roughly one review each. **Nothing about a representative market:** the sample's provenance was not recorded, so the figures describe this corpus and the method, not the category. And nothing about today — the data ends in 2012.